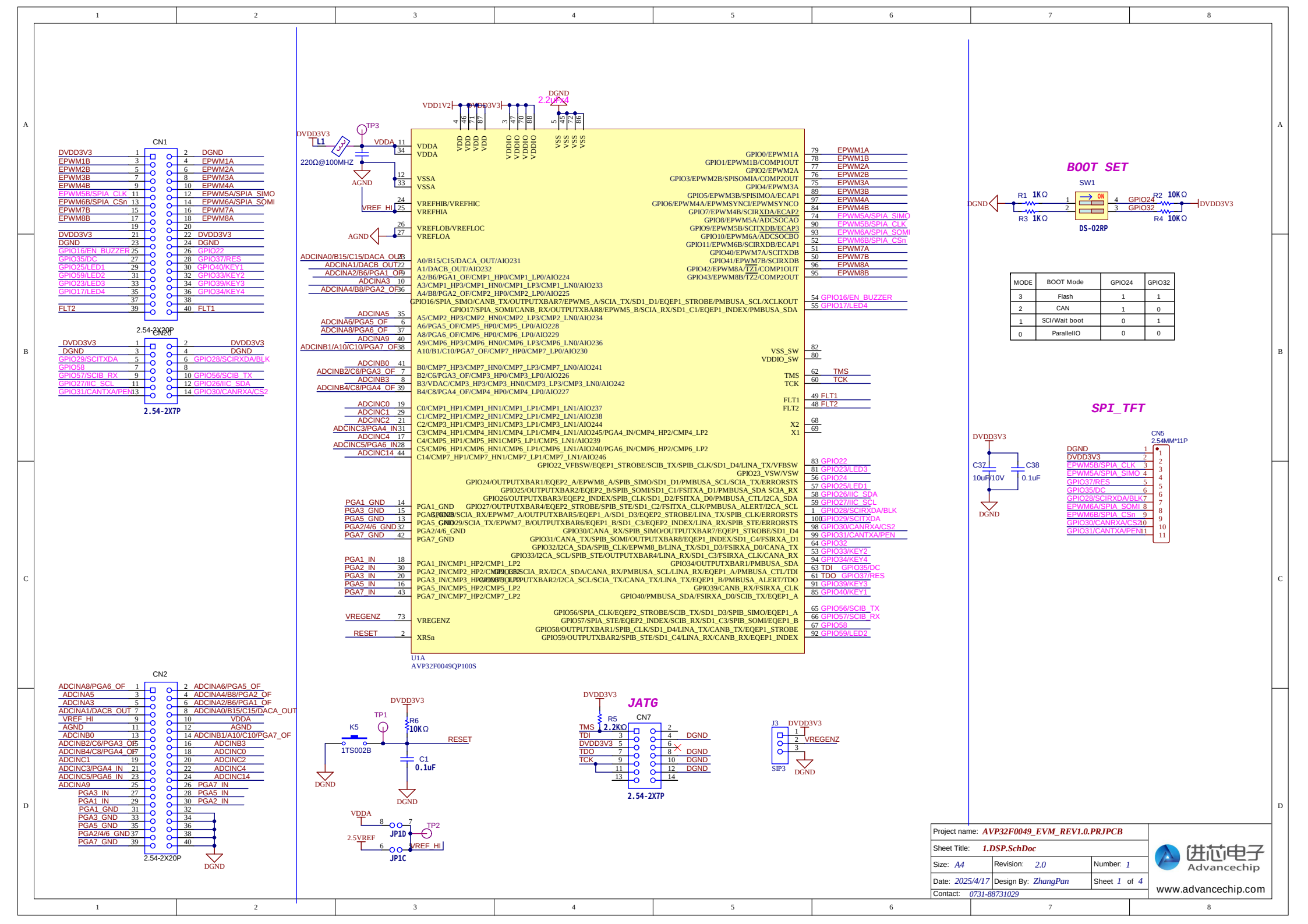
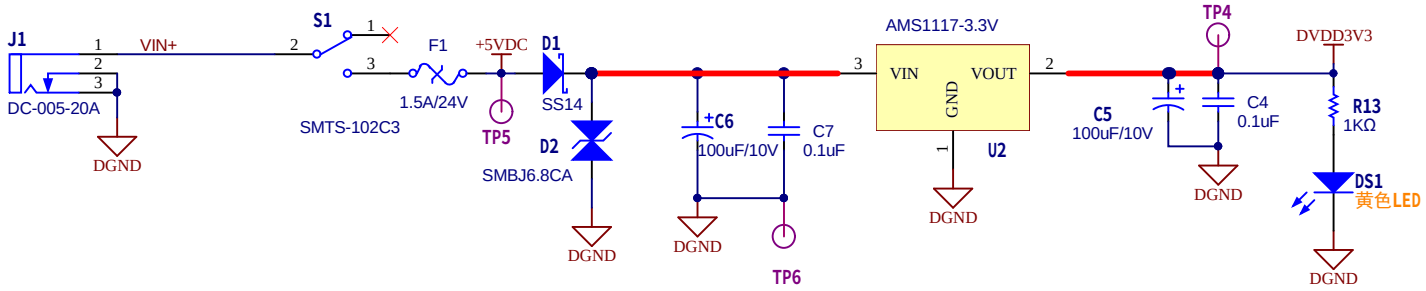
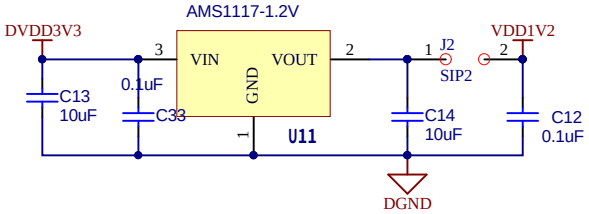
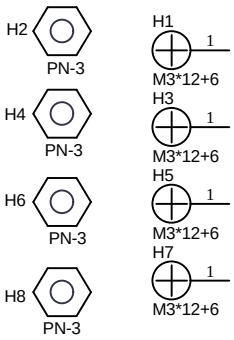
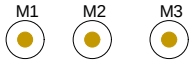

CN1
DVDD3V3 1
EPWM1B 3
EPWM2B 5
EPWM3B 7
EPWM4B 9
EPWM5B/SPIA_CLK 11
EPWM6B/SPIA_CSn 13
EPWM7B 15
EPWM8B 17
19
DVDD3V3 21
DGND 23
GPIO16/EN_BUZZER 25
GPIO35/DC 27
GPIO25/LED1 29
GPIO59/LED2 31
GPIO23/LED3 33
GPIO17/LED4 35
37
FLT2 39
2 DGND
4 EPWM1A
6 EPWM2A
8 EPWM3A
10 EPWM4A
12 EPWM5A/SPIA_SIMO
14 EPWM6A/SPIA_SOMI
16 EPWM7A
18 EPWM8A
20
22 DVDD3V3
24 DGND
26 GPIO22
28 GPIO37/RES
30 GPIO40/KEY1
32 GPIO33/KEY2
34 GPIO39/KEY3
36 GPIO34/KEY4
38
40 FLT1
2.54-2X20P
CN20
DVDD3V3 1
DGND 3
GPIO29/SCITXDA 5
GPIO58 7
GPIO57/SCIB_RX 9
GPIO27/IIC_SCL 11
GPIO31/CANTXA/PEN 13
2 DVDD3V3
4 DGND
6 GPIO28/SCIRXDA/BLK
8
10 GPIO56/SCIB_TX
12 GPIO26/IIC_SDA
14 GPIO30/CANRXA/CS2
2.54-2X7P
CN2
ADCINA8/PGA6_OF 1
ADCINA5 3
ADCINA3 5
ADCINA1/DACB_OUT 7
VREF_HI 9
AGND 11
ADCINB0 13
ADCINB2/C6/PGA3_OF 15
ADCINB4/C8/PGA4_OF 17
ADCINC1 19
ADCINC3/PGA4_IN 21
ADCINC5/PGA6_IN 23
ADCINA9 25
PGA3_IN 27
PGA1_IN 29
PGA1_GND 31
PGA3_GND 33
PGA5_GND 35
PGA2/4/6_GND 37
PGA7_GND 39
2 ADCINA6/PGA5_OF
4 ADCINA4/B8/PGA2_OF
6 ADCINA2/B6/PGA1_OF
8 ADCINA0/B15/C15/DACA_OUT
10 VDDA
12 AGND
14 ADCINB1/A10/C10/PGA7_OF
16 ADCINB3
18 ADCINC0
20 ADCINC2
22 ADCINC4
24 ADCINC14
26 PGA7_IN
28 PGA5_IN
30 PGA2_IN
32
34
36
38
40
2.54-2X20P
DGND
VDD1V2
DVDD3V3
DGND
2.2UFX4
4 46 71 87
3 47 70 88
5 45 72 86
VDD VDD VDD VDD
VDDIO VDDIO VDDIO VDDIO
VSS VSS VSS VSS
TP3
DVDD3V3
L1
220Ω@100MHZ
VDDA
AGND
VREF_HI
VDDA 11
VDDA 34
VSSA 12
VSSA 33
VREFHIB/VREFHIC 24
VREFHIA 25
VREFLOB/VREFLOC 26
VREFLOA 27
ADCINA0/B15/C15/DACA_OUT 8
ADCINA1/DACB_OUT 22
ADCINA2/B6/PGA1_OF 9
ADCINA3 10
ADCINA4/B8/PGA2_OF 36
ADCINA5 35
ADCINA6/PGA5_OF 6
ADCINA8/PGA6_OF 37
ADCINA9 40
ADCINB1/A10/C10/PGA7_OF 38
ADCINB0 41
ADCINB2/C6/PGA3_OF 7
ADCINB3 8
ADCINB4/C8/PGA4_OF 39
ADCINC0 19
ADCINC1 29
ADCINC2 21
ADCINC3/PGA4_IN 31
ADCINC4 17
ADCINC5/PGA6_IN 28
ADCINC14 44
PGA1_GND 14
PGA3_GND 15
PGA5_GND 13
PGA2/4/6_GND 32
PGA7_GND 42
PGA1_IN 18
PGA2_IN 30
PGA3_IN 20
PGA5_IN 16
PGA7_IN 43
VREGENZ 73
RESET 2
A0/B15/C15/DACA_OUT/AIO231
A1/DACB_OUT/AIO232
A2/B6/PGA1_OF/CMP1_HP0/CMP1_LP0/AIO224
A3/CMP1_HP3/CMP1_HN0/CMP1_LP3/CMP1_LN0/AIO233
A4/B8/PGA2_OF/CMP2_HP0/CMP2_LP0/AIO225
GPIO16/SPIA_SIMO/CANB_TX/OUTPUTXBAR7/EPWM5_A/SCIA_TX/SD1_D1/EQEP1_STROBE/PMBUSA_SCL/XCLKOUT
GPIO17/SPIA_SOMI/CANB_RX/OUTPUTXBAR8/EPWM5_B/SCIA_RX/SD1_C1/EQEP1_INDEX/PMBUSA_SDA
A5/CMP2_HP3/CMP2_HN0/CMP2_LP3/CMP2_LN0/AIO234
A6/PGA5_OF/CMP5_HP0/CMP5_LP0/AIO228
A8/PGA6_OF/CMP6_HP0/CMP6_LP0/AIO229
A9/CMP6_HP3/CMP6_HN0/CMP6_LP3/CMP6_LN0/AIO236
A10/B1/C10/PGA7_OF/CMP7_HP0/CMP7_LP0/AIO230
B0/CMP7_HP3/CMP7_HN0/CMP7_LP3/CMP7_LN0/AIO241
B2/C6/PGA3_OF/CMP3_HP0/CMP3_LP0/AIO226
B3/VDAC/CMP3_HP3/CMP3_HN0/CMP3_LP3/CMP3_LN0/AIO242
B4/C8/PGA4_OF/CMP4_HP0/CMP4_LP0/AIO227
C0/CMP1_HP1/CMP1_HN1/CMP1_LP1/CMP1_LN1/AIO237
C1/CMP2_HP1/CMP2_HN1/CMP2_LP1/CMP2_LN1/AIO238
C2/CMP3_HP1/CMP3_HN1/CMP3_LP1/CMP3_LN1/AIO244
C3/CMP4_HP1/CMP4_HN1/CMP4_LP1/CMP4_LN1/AIO245/PGA4_IN/CMP4_HP2/CMP4_LP2
C4/CMP5_HP1/CMP5_HN1CMP5_LP1/CMP5_LN1/AIO239
C5/CMP6_HP1/CMP6_HN1/CMP6_LP1/CMP6_LN1/AIO240/PGA6_IN/CMP6_HP2/CMP6_LP2
C14/CMP7_HP1/CMP7_HN1/CMP7_LP1/CMP7_LN1/AIO246
PGA1_IN/CMP1_HP2/CMP1_LP2
PGA2_IN/CMP2_HP2/CMP2_LP2
PGA3_IN/CMP3_HP2/CMP3_LP2
PGA5_IN/CMP5_HP2/CMP5_LP2
PGA7_IN/CMP7_HP2/CMP7_LP2
VREGENZ
XRSn
GPIO0/EPWM1A 79
GPIO1/EPWM1B/COMP1OUT 78
GPIO2/EPWM2A 77
GPIO3/EPWM2B/SPISOMIA/COMP2OUT 76
GPIO4/EPWM3A 75
GPIO5/EPWM3B/SPISIMOA/ECAP1 89
GPIO6/EPWM4A/EPWMSYNCI/EPWMSYNCO 97
GPIO7/EPWM4B/SCIRXDA/ECAP2 84
GPIO8/EPWM5A/ADCSOCAO 74
GPIO9/EPWM5B/SCITXDB/ECAP3 90
GPIO10/EPWM6A/ADCSOCBO 93
GPIO11/EPWM6B/SCIRXDB/ECAP1 52
GPIO40/EPWM7A/SCITXDB 51
GPIO41/EPWM7B/SCIRXDB 50
GPIO42/EPWM8A/TZ1/COMP1OUT 96
GPIO43/EPWM8B/TZ2/COMP2OUT 95
54 GPIO16/EN_BUZZER
55 GPIO17/LED4
VSS_SW 82
VDDIO_SW 80
TMS 62
TCK 60
FLT1 49
FLT2 48
X2 68
X1 69
GPIO22 83
GPIO23/LED3 81
GPIO24 56
GPIO25/LED1 57
GPIO26/IIC_SDA 58
GPIO27/IIC_SCL 59
GPIO28/SCIRXDA/BLK 1
GPIO29/SCITXDA 100
GPIO30/CANRXA/CS2 98
GPIO31/CANTXA/PEN 99
GPIO32 64
GPIO33/KEY2 53
GPIO34/KEY4 94
TDI GPIO35/DC 63
TDO GPIO37/RES 61
GPIO39/KEY3 91
GPIO40/KEY1 85
GPIO56/SCIB_TX 65
GPIO57/SCIB_RX 66
GPIO58 67
GPIO59/LED2 92
U1A
AVP32F0049QP100S
DVDD3V3
TP1
K5
1TS002B
R6
10KΩ
RESET
C1
0.1uF
DGND
VDDA
TP2
2.5VREF
JP1D
JP1C
VREF_HI
DVDD3V3
JATG
R5
2.2KΩ
CN7
TMS 1
TDI 3
DVDD3V3 5
TDO 7
TCK 9
2 4 6 8 10 12 14
DGND
2.54-2X7P
J3 DVDD3V3
1 2 3
VREGENZ
SIP3
DGND
BOOT SET
SW1
DGND
R1 1KΩ
R3 1KΩ
ON
1 2 4 3
GPIO24
GPIO32
R2 10KΩ
R4 10KΩ
DVDD3V3
DS-02RP
MODE BOOT Mode GPIO24 GPIO32
3 Flash 1 1
2 CAN 1 0
1 SCI/Wait boot 0 1
0 ParallelIO 0 0
SPI_TFT
DVDD3V3
C37
10uF/10V
C38
0.1uF
DGND
CN5
2.54MM*11P
DGND 1
DVDD3V3 2
EPWM5B/SPIA_CLK 3
EPWM5A/SPIA_SIMO 4
GPIO37/RES 5
GPIO35/DC 6
GPIO28/SCIRXDA/BLK 7
EPWM6A/SPIA_SOMI 8
EPWM6B/SPIA_CSn 9
GPIO30/CANRXA/CS2 10
GPIO31/CANTXA/PEN 11
Project name: AVP32F0049_EVM_REV1.0.PRJPCB
Sheet Title: 1.DSP.SchDoc
Size: A4
Revision: 2.0
Number: 1
Date: 2025/4/17
Design By: ZhangPan
Sheet 1 of 4
Contact: 0731-88731029
进芯电子
Advancechip
www.advancechip.com

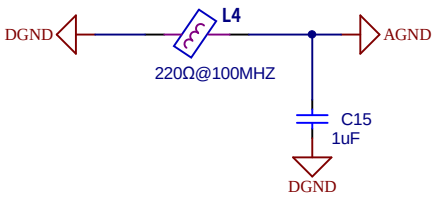


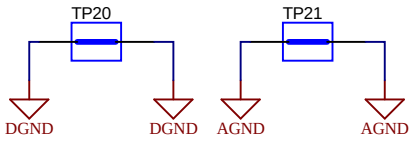




PCB
LOGO



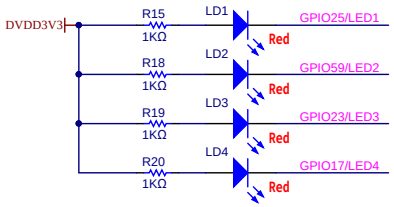




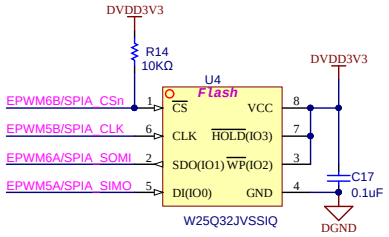
Project name: *AVP32F0049_EVM_REV1.0.PRJPCB*		
Sheet Title: *2.Power.SchDoc*		
Size: *A4*	Revision: *1.0*	Number: *2*
Date: *2025/4/17*	Design By: *DaiJun*	Sheet *2* of *4*
Contact: *0731-88731029*		

进芯电子 Advancechip
www.advancechip.com

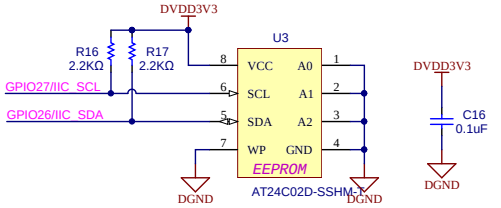
LED X 4



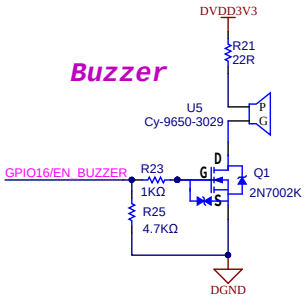
Flash



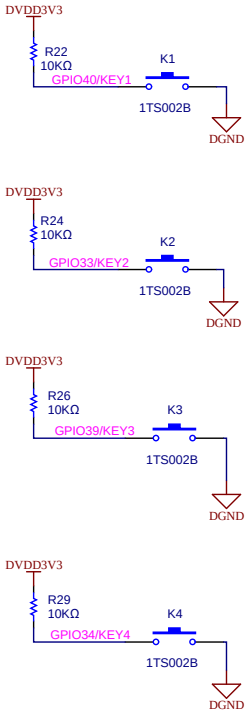
EEPROM



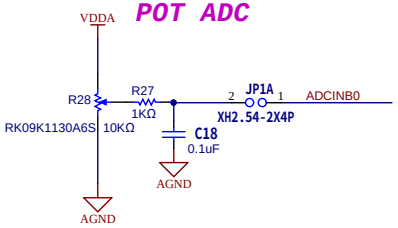
Buzzer



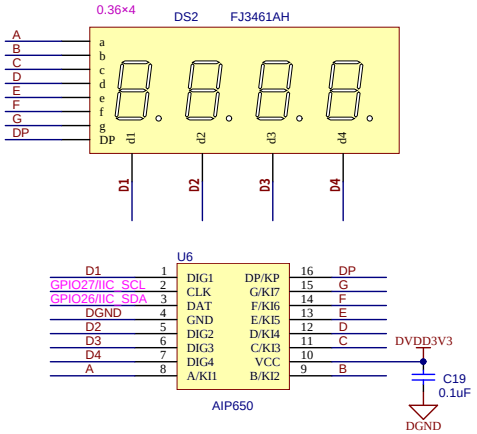
Switch X 4



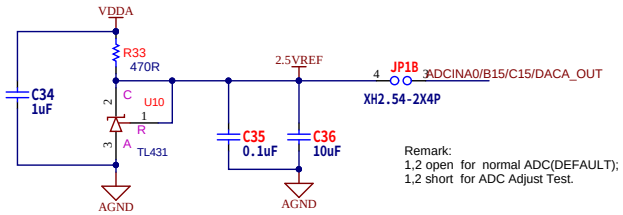
POT ADC



DisPlay



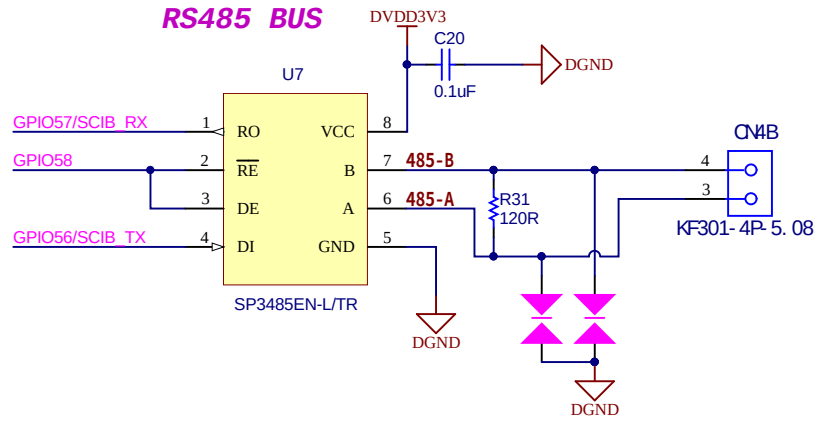
2.5VREF

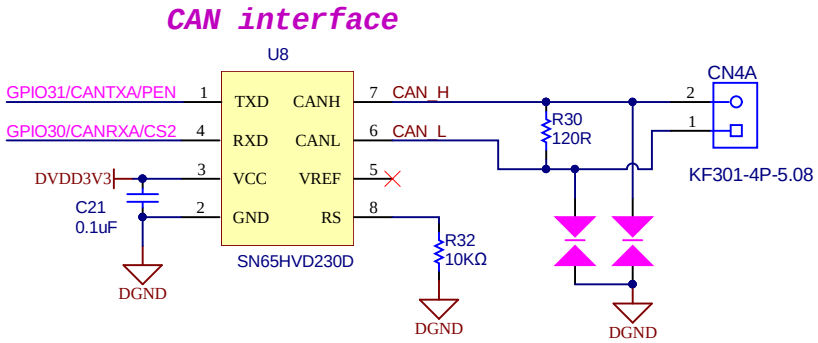


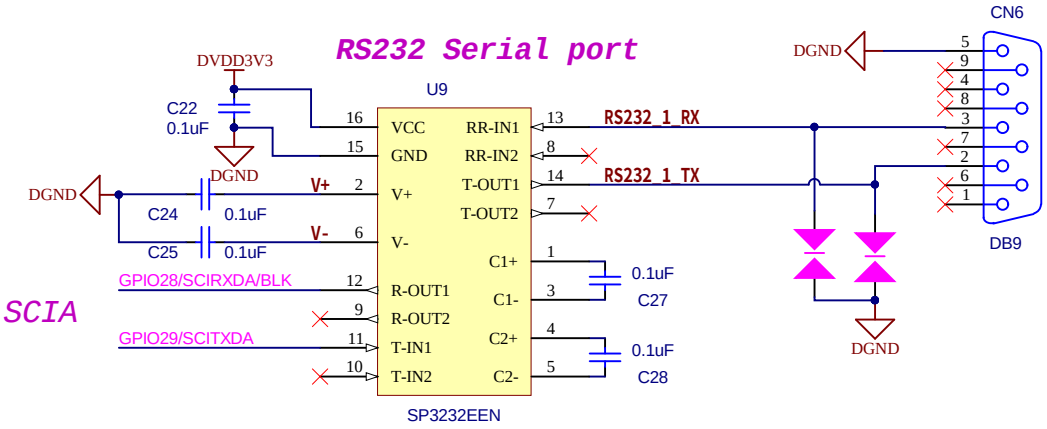
Remark:
1,2 open for normal ADC(DEFAULT);
1,2 short for ADC Adjust Test.

Project name: AVP32F0049_EVM_REV1.0.PRJPCB		
Sheet Title: 3.uC Peripheral.SchDoc		
Size: A4	Revision: 2.0	Number: 3
Date: 2025/4/17	Design By: ZhangPan	Sheet 3 of 4
Contact: 0731-88731029		

进芯电子 Advancechip
www.advancechip.com







Project name: *AVP32F0049_EVM_REV1.0.PRJPCB*		
Sheet Title: *4.Communication Interface.SchDoc*		
Size: *A3*	Revision: *1.0*	Number: *4*
Date: *2025/4/17*	Design By: *DaiJun*	Sheet *4* of *4*
Contact: *0731-88731029*		

进芯电子 Advancechip
www.advancechip.com